



LINUX Interview Style Q&A with Practical Context DAY 11

Linux Filesystem Basics – Interview-Style Q&A

1.How do you create an empty file?

Command: touch filename

Explanation: Creates a new, empty file named filename. If the file already exists, touch updates its access and modification timestamps.

2.How do you create a new directory?

Command: mkdir directory_name

Explanation: Creates a directory named directory_name in the current working directory. Use -p to create nested directories.

3.How do you list all files in a directory?

Command: ls

Explanation: Displays the names of files and directories in the current directory (excluding hidden files).

4.How do you list all files, including hidden ones?

Command: ls -a

Explanation: Lists all files, including hidden files (those starting with a .).

5.How do you display the current working directory?

Command: pwd

Explanation: Prints the absolute path of your current directory.

6.How do you move or rename a file?

Command: mv source_file destination_file

Explanation: Moves a file or directory to a new location or renames it if destination_file is a new name in the same directory.

7.How do you copy a file?

Command: cp source_file destination_file

Explanation: Copies source_file to destination_file. Use -r for recursive directory copy and -p to preserve timestamps.

8.What command is used to delete a file?

Command: rm filename

Explanation: Deletes the file filename. Be cautious, as this action is irreversible.

9.How do you delete a directory and its contents?

Command: rm -r directory_name

Explanation: Recursively deletes a directory and all files/subdirectories within it. Use with care.

10.How do you display the contents of a file?

Command: cat filename

Explanation: Outputs the content of the file to the terminal. Ideal for viewing short files.

11.How do you display the first 10 lines of a file?

Command: `head filename`

Explanation: By default, prints the first 10 lines. Use `head -n number filename` to customize line count.

12.How do you display the last 10 lines of a file?

Command: `tail filename`

Explanation: By default, prints the last 10 lines. Use `tail -n number filename` to change line count.

13.How do you check free disk space?

Command: `df -h`

Explanation: Displays disk space usage for all mounted filesystems in human-readable format (e.g., GB/MB).

14.How do you display disk usage of a directory?

Command: `du -sh directory_name`

Explanation: Shows the total disk space used by the specified directory in a human-readable format.

15.How do you search for a file by name in a directory?

Command: `find /path/to/directory -name filename`

Explanation: Searches recursively for a file named filename starting from the specified path.

16.How do you compress a file using gzip?

Command: `gzip filename`

Explanation: Compresses filename and replaces it with filename.gz.

17.How do you decompress a gzip file?

Command: `gunzip filename.gz`

Explanation: Decompresses filename.gz, restoring the original file.

18.How do you archive multiple files using tar?

Command: `tar -czvf archive.tar.gz file1 file2 ...`

Explanation: Creates a compressed archive (.tar.gz) from the listed files.
c: create, z: gzip compression, v: verbose, f: filename.

19.How do you extract a tar.gz archive?

Command: `tar -xzvf archive.tar.gz`

Explanation: Extracts files from a .tar.gz archive.

20.How do you mount a filesystem?

Command: `mount device_name mount_point`

Explanation: Mounts the specified device (like /dev/sdb1) to the directory path mount_point.
Requires root privileges.

21.How do you unmount a filesystem?

Command: `umount mount_point`

Explanation: Unmounts the mounted filesystem. Ensure no files are in use from the mount point.

22.How do you create a symbolic link?

Command: `ln -s target_file link_name`

Explanation: Creates a symbolic (soft) link pointing to `target_file` with the name `link_name`.

23.How do you view file types in a directory?

Command: `file filename`

Explanation: Displays the file type (text, binary, directory, etc.) based on its content, not extension.

24.How do you remove an empty directory?

Command: `rmdir directory_name`

Explanation: Deletes a directory only if it is empty.

25.How do you check for file existence in a script?

Command: `test -f filename`

Explanation: Returns true if `filename` exists and is a regular file. Commonly used in shell scripts.

Alternative syntax: `[ -f filename ]`

Linux Permissions & ACLs – Interview-Style Questions

26.A script fails to run due to permission issues. How do you modify its permissions?

A: Use `chmod 755 script.sh`. This gives the owner full permissions and read/execute to others. Alternatively, `chmod +x script.sh` makes it executable without altering other permissions.

27.You need to change ownership of a deployment artifact to a specific user and group. What command do you use?

A: `chown devops:deployments artifact.zip` changes both the owner and group, ensuring proper access control post-deployment.

28.How do you verify file permissions and ownership in a directory when debugging access issues?

A: Use `ls -l`. This shows detailed information, including permission bits (rwx), ownership (user and group), file size, and last modified time.

29.You've deployed a web application and need to make the entire directory readable and executable by all users. How?

A: Use `chmod -R 755 /var/www/html`. This ensures the web server can access all files and directories within the path.

30. A teammate used `chmod 777` on a directory. What does this mean and is it secure?

A: `chmod 777` grants read, write, and execute permissions to everyone, which is insecure in most cases. Any user can modify or delete the contents. It's suitable only in controlled, temporary environments.

31. Q: How do you check if a file has extended ACLs applied?

A: Use `getfacl filename`. It displays traditional and extended ACL entries. Extended ACLs override standard user/group/other permissions.

32. Q: In a shared folder, how do you ensure that new files created inside always allow a specific user read/write access?

A: Set a default ACL using:

```
setfacl -d -m u:john:rw shared_folder/
```

This automatically applies to all newly created files/directories.

33. Q: You've created a script but can't run it. How do you make it executable?

A: Use `chmod +x deploy.sh`. This adds execute permission for the user, making it runnable as `./deploy.sh`.

34. Q: What is the purpose of the sticky bit on directories like `/tmp`?

A: The sticky bit prevents users from deleting files they don't own, even if the directory is world-writable. It's a security measure for shared environments.

35. Q: How do you apply the sticky bit to a shared directory to prevent unauthorized file deletion?

A: Run `chmod +t /shared_folder`. This ensures only file owners or root can delete files, commonly used for directories like `/tmp`.

Process Management – Interview-Style Q&A

36. You're troubleshooting a system and need to see what processes are running. What command would you use?

A: I'd use `ps -aux` to list all running processes, including those without a controlling terminal. It shows details like PID, user, CPU/memory usage, and the command running the process.

37. A service is unresponsive. You have the PID. How do you stop it?

A: I'd use `kill PID` to send a default `SIGTERM` signal to gracefully stop the process.

38. You want to kill a process but only know its name. How do you find its PID?

A: I'd use `pgrep process_name`, which returns the PID(s) of processes matching the given name.

39. How would you continuously monitor system resource usage by processes?

A: I'd use `top` for real-time monitoring, or `htop` if available, for a more interactive and color-coded interface.

40. What's the difference between `kill -15` and `kill -9`? When would you use each?

A:

- `kill -15` sends a `SIGTERM` for a graceful shutdown (preferred).
- `kill -9` sends a `SIGKILL` for a forceful termination (used when the process doesn't respond to `SIGTERM`).

41. Q: A process doesn't shut down with a regular kill. What do you do next?

A: I'd run `kill -9 PID` to forcefully terminate the process using `SIGKILL`.

42. How do you run a long-running job in the background from the shell?

A: By appending & to the command. For example:

```
./backup.sh &
```

This allows me to keep using the terminal while the job runs in the background.

43. You've sent a job to the background. How do you bring it back to the foreground?

A: I'd use fg %job_number. If it's the only background job, simply fg will bring it to the foreground.

44. How do you monitor resource usage like CPU and memory for a specific process over time?

A: I'd use pidstat -p <PID>, which gives detailed per-process statistics.

45. How do you view logs for a specific systemd service, like nginx?

A: I'd use:

```
journalctl -u nginx
```

To see recent logs, I might add --since "1 hour ago".

46. You often use a long command and want to shorten it. How do you do that?

A: I'd create an alias using:

```
alias shortname='long_command_here'
```

For example, alias ll='ls -aF'.

47. How do you remove or temporarily disable an alias you've set?

A: I'd use unalias alias_name. For example:

```
unalias ll
```

48. How do you run a command as another user without switching sessions?

A: I'd use sudo -u username command. For example, to connect to a PostgreSQL DB as the postgres user:

```
sudo -u postgres psql
```

49. How do you list all environment variables currently set in your shell?

A: I'd use printenv or env. Both list the current environment variables.

50. How do you set and export an environment variable so it's accessible to child processes or scripts?

A: I'd use the export command. For example:

```
export APP_ENV=production
```